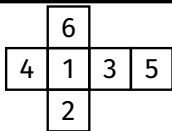
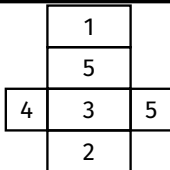


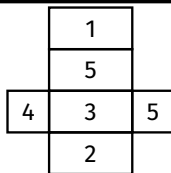
Push Tasks - Nets



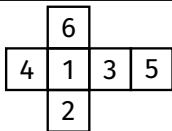
- 1. Invalid arrangements?**
e.g. 4 in a row with 2 on same side — overlaps



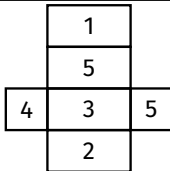
- 3. 3 pairs face areas?**
15, 10, 6 each x 2



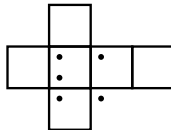
- 5. Max SA for $V=96$?**
 $1 \times 1 \times 96 = 386 \text{ cm}^2$



- 2. Why some fail?**
Adjacent squares overlap

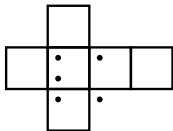


- 4. Min SA for $V=96$?**



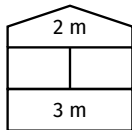
- 6. Opposite sums to 7?**
Standard: 1-6 2-5 3-4

Push Tasks - Nets



7. Different die net?

Any valid cube net works



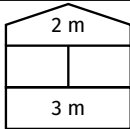
9. Different net?

Swap triangle positions



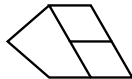
11. Rectangular pyramid?

Net: 1 rect + 4 triangles (2 pairs)



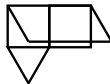
8. Triangles isosceles?

Two equal sides by



10. Triangles congruent?

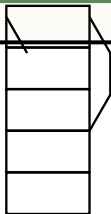
Yes — same base and slant height



12. Right triangle prism?

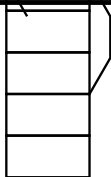
Rectangles + 2 right triangles

Push Tasks - Nets



13. Distinct nets?

Many — 2 hexagons + 6



14. Hex prism SA?



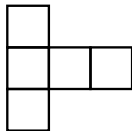
15. Tetrahedron nets?

2 distinct nets



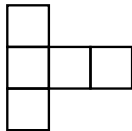
16. Why 4 triangles?

4 faces, all triangles



17. Lid positions?

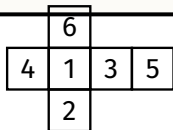
Depends on shape



18. $V=12$ no top SA?

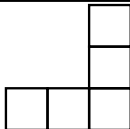
Depends on dimensions

Push Tasks - Nets



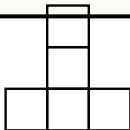
19. Shortest surface path?

Unfold net, draw straight line



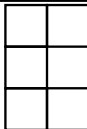
20. L as cube net?

Only if 6th square added



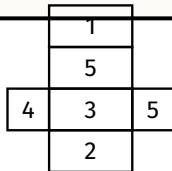
21. 2 non-box pentominoes?

I-pentomino (straight) and X-pentomino



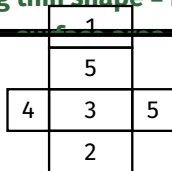
22. All open box nets?

11 pentominoes can form



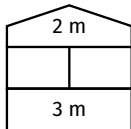
23. 1x1x96 max paper?

Long thin shape = huge



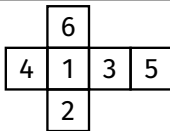
24. 4x6x4 min paper proof?

Push Tasks - Nets



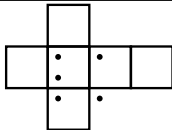
25. Ribbon on faces?

3.8 m on faces, 0.2 m knot



27. Shortest path?

**Draw net, connect start
to end**



26. Min colours?

3 (opposite faces same)